Evaluating Out-of-Distribution Robustness of TabPFN on Tabular Data

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***Abstract*—Out-of-distribution (OOD) generalization is a critical challenge in machine learning, especially for tabular data where real-world distributions often differ from training data. TabPFN is a recent transformer-based model that achieves strong performance on small tabular classification tasks by training only on synthetic data. However, its ability to generalize to OOD scenarios in real-world datasets has not been fully explored. In this work, the evaluation of the pre-trained TabPFN model is extended to the UCI Adult dataset, focusing on OOD splits based on education, age, and gender. TabPFN's performance is compared with traditional models such as Random Forest, XGBoost, and LinearSVC. Experiments show that TabPFN consistently achieves higher accuracy and F1 scores on OOD samples across different demographic splits. Visualizations and statistical analysis are provided to support these findings. This study demonstrates the practical value of TabPFN for robust tabular classification in the presence of distribution shifts, and highlights its potential for real-world applications where fairness and reliability are important.**

Keywords—tabular data, classification, out-of-distribution generalization, tabPFN, demographic shift

I. Introduction

Tabular data is widely used in real-world machine learning applications, including finance, healthcare, and social sciences. While deep learning has achieved remarkable results in fields such as computer vision and natural language processing, traditional models such as gradient-boosted trees and random forests are still widely used for tabular data due to their robustness and efficiency [1,2]. A major challenge in this area is out-of-distribution (OOD) generalization. OOD occurs when the data seen during deployment is different from the data used for training. This difference is called a distribution shift. OOD analysis is important because models often face new or changing data in real-world use. If OOD generalization is not addressed, models may make poor predictions on new data. For example, a model trained on one group may not work well for another group. This can cause errors, unreliable results, or even unfair decisions. In sensitive areas like healthcare or finance, this can lead to serious negative impacts for individuals or groups.

Recent research has introduced TabPFN, a transformer-based model trained entirely on synthetic data, which achieves strong results on small tabular classification tasks [3]. However, the OOD generalization ability of TabPFN on real-world datasets has not been explored. In many practical scenarios, models must perform well not only on in-distribution data but also on subgroups or conditions that were not present during training. While recent work has explored TabPFN's ability to handle temporal distribution shifts [11], its robustness to demographic distribution shifts, which are critical for fairness and ethical AI deployment, remains unexplored. This paper addresses this gap by evaluating the pre-trained TabPFN model on the UCI Adult dataset, focusing on OOD splits based on education, age, and gender. Unlike prior work on temporal shifts, this evaluation examines demographic distribution shifts without model modification, directly addressing fairness concerns through systematic statistical analysis. The performance of TabPFN is compared with established models such as Random Forest, XGBoost, and LinearSVC. The main contributions of this work are as follows:

- A systematic evaluation of TabPFN's zero-shot OOD generalization on the UCI Adult dataset using realistic demographic splits. This is the first evaluation of TabPFN on demographic distribution shifts, addressing fairness-critical scenarios not explored in prior work.
- A comparison of TabPFN with traditional machine learning models under demographic OOD conditions, providing statistical evidence that the observed performance differences are significant and not due to random chance.
- Visual and statistical analysis to highlight the strengths and limitations of TabPFN in OOD scenarios.

II. Related Work

Machine learning on tabular data has traditionally relied on models such as decision trees, random forests, and gradient-boosted machines due to their strong performance and ease of use [1,2]. While deep learning has transformed fields like image and text analysis, its impact on tabular data has been limited. Recent surveys highlight that deep neural networks often do not outperform tree-based models on standard tabular benchmarks [1,2].

Several approaches have been proposed to improve deep learning for tabular data, including specialized architectures and regularization techniques [1,2,7,8]. However, these methods often require careful tuning and may not generalize well to new datasets [1]. The challenge of out-of-distribution (OOD) generalization is especially important, as real-world data distributions can shift due to demographic, temporal, or other factors. Recent benchmarking efforts have systematically evaluated distribution shift in tabular data across diverse domains including finance, education, and healthcare, revealing that OOD performance often degrades substantially compared to in-distribution accuracy [12]. These findings underscore the need for robust models that can maintain performance under distributional changes.

TabPFN is a recent model that addresses tabular classification by training a transformer on millions of synthetic datasets generated from a carefully designed prior [3]. This approach enables the model to perform in-context learning and make predictions without further training or hyperparameter tuning. This meta-learning approach allows

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the model to generalize by learning patterns across diverse data distributions rather than memorizing task-specific features [3,9]. The original TabPFN paper demonstrates strong results on small, numerical datasets from the OpenML-CC18 benchmark, but does not focus on OOD generalization in real-world demographic splits.

Recent work has extended the capabilities of meta-learning models for OOD scenarios. Drift-Resilient TabPFN demonstrates the ability to handle temporal distribution shifts by incorporating structural causal models into the training prior, achieving significant improvements over traditional methods on real-world drift scenarios [11]. Alternative approaches based on contrastive representation learning have also shown promise for OOD detection in tabular data, though these methods often require specialized hardware [13]. However, the robustness of TabPFN to demographic distribution shifts remains underexplored.

The UCI Adult dataset is a widely used benchmark for income prediction and fairness analysis. It contains demographic features that allow for realistic OOD splits based on education, age, and gender. Evaluating models on such splits provides insight into their robustness and fairness in practical scenarios [4].

This work extends the evaluation of TabPFN by testing its OOD generalization on the UCI Adult dataset and comparing it with established baselines. The results provide new evidence on the strengths and limitations of transformer-based models for tabular OOD generalization.

III. Methodology

A. Dataset and Preprocessing

The UCI Adult dataset [4] was used for all experiments. This dataset contains 45,222 records after preprocessing. Each record represents an individual described by 14 features, including both numerical and categorical variables such as age, education, occupation, and gender. The target variable is binary, indicating whether the individual's annual income is greater than $50,000.

Data preprocessing involved several steps. All rows with missing values were removed. Categorical features were converted to numeric values using label encoding. The income column was mapped to 0 for incomes less than or equal to $50,000 and 1 for incomes above this threshold. The original training and test splits from the UCI repository were combined to create a single dataset. This allows for flexible and consistent splitting for OOD experiments.

B. Out-of-Distribution (OOD) Split Strategies

To study OOD generalization, three types of splits were created based on demographic attributes: education, age, and gender. For each split, the entire dataset was divided into in-distribution and OOD groups based on specific feature thresholds, as shown in Table I. This setup allows models to be trained on one demographic subset and evaluated on a systematically different subgroup, simulating a real-world distribution shift.

TABLE I. Criteria for In-Distribution and OOD Groups in Each Split

Split	In-Distribution	Out-of-Distribution (OOD)
Education	education-num <= 10	education-num > 10
Age	age <= 35	age > 35
Gender	male (sex = 0 after label encoding)	female (sex = 1 after label encoding)

TABLE II. In-Distribution and OOD Sample Counts for Education, Age, and Gender Splits (UCI Adult Dataset)

	Education	Age	Gender
Total	45222	45222	45222
In-Distribution Count	30343	20515	14695
In-Distribution %	67.10%	45.37%	32.50%
OOD Count	14879	24707	30527
OOD %	32.90%	54.63%	67.50%
In-Distribution Train	21240	14360	10286
In-Distribution Test	9103	6155	4409
OOD Test	1000	1000	1000

For each scenario, the in-distribution group was further split into a training set (70%) and a test set (30%) using a fixed random seed for reproducibility. The OOD group was reserved for testing only. For OOD evaluation, up to 1,000 samples were randomly selected from the OOD group for each split. Table II summarizes the exact counts for each split.

C. Model Training and Evaluation

Four different models are included in the evaluation: TabPFN, Random Forest, XGBoost, and LinearSVC. TabPFN is a transformer-based classifier that was trained entirely on synthetic data [3]. In this study, the pre-trained TabPFN model is used directly. No additional training or fine-tuning is performed on the UCI Adult dataset.

TABLE III. Summary of Training and Test Data Utilization for TabPFN and Baseline Models

Model	Trained on UCI Adult?	Training Data Used	Test Data Used
TabPFN	No	None (pre-trained only)	In-distribution test, OOD set
Random Forest	Yes	In-distribution training	In-distribution test, OOD set
XGBoost	Yes	In-distribution training	In-distribution test, OOD set
LinearSVC	Yes	In-distribution training	In-distribution test, OOD set

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TabPFN is evaluated in a zero-shot setting on the UCI Adult dataset, without any fine-tuning or parameter updates. Following the TabPFN methodology [3], the model performs in-context learning (ICL) by using in-distribution training samples as labeled examples during inference. It is only used to make predictions on the in-distribution test set and the OOD test set. It never sees any part of the UCI Adult data during training. Random Forest, XGBoost, and LinearSVC are traditional machine learning models. These models were trained only on the in-distribution training set for each split. For example, in the education split, the model was trained only on the 21,240 individuals with education-num <= 10. The remaining 9,103 in-distribution samples were used as the in-distribution test set. The OOD group (e.g., education-num > 10) was never used for training. It was used only for testing OOD generalization. The same procedure was followed for the age and gender splits. Table III summarizes the training and testing data for each model.

All experiments were implemented in Python using standard libraries such as pandas, scikit-learn, xgboost, and the official TabPFN package. Random seeds were set for all sampling and model training steps to ensure reproducibility.

D. Evaluation Metrics and Statistical Analysis

Model performance was measured using accuracy and F1 score. Accuracy is the proportion of correct predictions, while F1 score balances precision and recall, making it suitable for imbalanced classes. To assess the statistical significance of differences between models, paired t-tests and Wilcoxon signed-rank tests were performed on the per-sample accuracy and F1 scores. The results were summarized in tables and visualized in the Results and Discussion section. Statistical test results were also reported to support the findings.

E. Fairness and Limitations of the Comparison

It is important to note the differences in how the models were used and the group sizes for each split. TabPFN was evaluated in a zero-shot setting, using the pre-trained model without any additional training on the UCI Adult dataset. In contrast, the baseline models were trained on the in-distribution training split. For each OOD scenario, the in-distribution and OOD group sizes varied due to the underlying data distribution (Fig. 1, Fig. 2, Fig. 3).

While this reflects real-world demographic shifts, it also means that the amount of training data available to the models differed by scenario. To ensure a fair comparison between models within each scenario, all models were trained and tested on the same splits. However, results should not be directly compared across different OOD scenarios due to these differences. In this study, all model comparisons are made within the same OOD split. No direct comparisons are made across different OOD scenarios.

IV. Results and Discussion

A. Class Distribution Shifts in OOD Splits

To rigorously evaluate out-of-distribution (OOD) generalization, three OOD scenarios were constructed based on education, age, and gender, following established practices for fairness and robustness analysis in tabular data [2,4]. The class distribution plots for each split (Fig. 1, Fig. 2, Fig. 3) reveal substantial differences between the in-distribution (ID) and OOD groups.

For the Education split (Fig. 1), the OOD group (education-num > 10) contains more samples in both income classes compared to the ID group, but the class balance is notably different. The OOD group is more evenly distributed, while the ID group is skewed towards the lower income class (<=50K). This shift creates a more challenging test for model generalization, as the model must adapt to a new class balance not seen during training.

In the Age split (Fig. 2), the OOD group (age > 35) is much larger than the ID group and contains a higher proportion of both income classes. The OOD group has a substantial increase in the number of samples with income <=50K, as well as a higher count of >50K compared to the ID group. This demographic shift reflects real-world scenarios where age distributions can change over time or across populations [4].

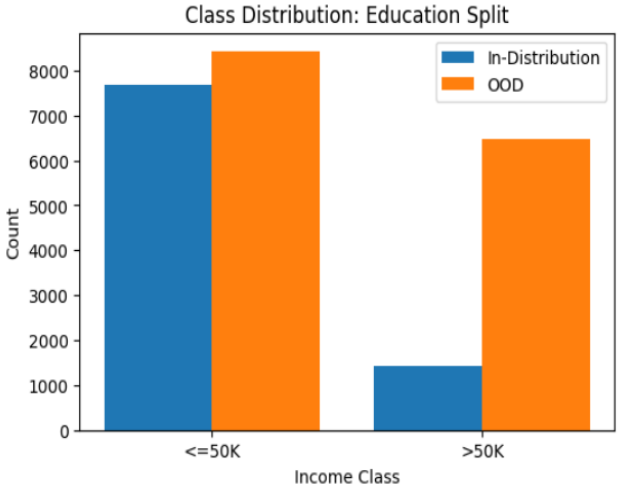


Fig. 1. Class distribution of income for the Education split.

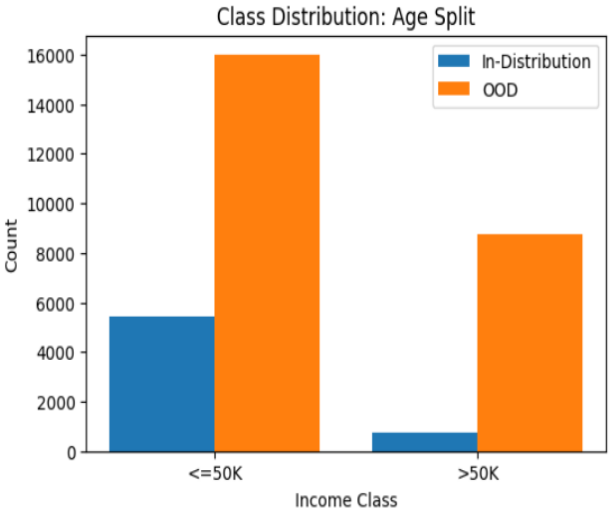


Fig. 2. Class distribution of income for the Age split.

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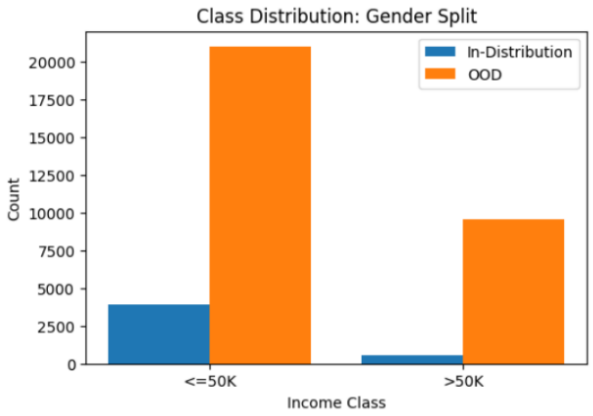


Fig. 3. Class distribution of income for the Gender split.

For the Gender split (Fig. 3), the OOD group (females) is characterized by a much higher count of samples with income <=50K, while the ID group (males) is more balanced between the two income classes. This highlights a significant demographic and class imbalance, which is a realistic and challenging setting for evaluating model robustness and fairness [2].

These observed shifts confirm that the OOD splits are non-trivial and present realistic challenges for model generalization. The models are required to make predictions on data distributions that differ substantially from those seen during training, which is a critical test for practical deployment [1,2].

B. *Model Performance: In-Distribution and OOD Evaluation*

TABLE IV. IN-DISTRIBUTION (ID) AND OUT-OF-DISTRIBUTION (OOD) ACCURACY AND F1 SCORES FOR ALL MODELS AND SPLITS

Split	Model	ID Accuracy	ID F1	OOD Accuracy	OOD F1
Education	TabPFN	0.880	0.513	0.800	0.724
	Random Forest	0.876	0.507	0.755	0.625
	XGBoost	0.885	0.533	0.768	0.660
	LinearSVC	0.853	0.231	0.669	0.379
Age	TabPFN	0.914	0.569	0.803	0.700
	Random Forest	0.911	0.561	0.789	0.649
	XGBoost	0.922	0.622	0.796	0.659
	LinearSVC	0.891	0.230	0.693	0.297
Gender	TabPFN	0.928	0.655	0.803	0.642
	Random Forest	0.926	0.641	0.746	0.401
	XGBoost	0.933	0.680	0.769	0.514
	LinearSVC	0.900	0.304	0.722	0.291

All models were evaluated on both in-distribution (ID) and out-of-distribution (OOD) data as shown in Table IV.

In the in-distribution (ID) setting, XGBoost achieves the highest or among the highest accuracy and F1 scores across all splits. This result is consistent with prior findings. Previous studies have shown that tree-based models are strong performers on tabular data [1,2]. Random Forest also performs well, and TabPFN is close behind. LinearSVC consistently underperforms, particularly in F1 score, indicating its limited suitability for this tabular classification task.

In the out-of-distribution (OOD) setting, TabPFN consistently achieves the highest or near-highest accuracy and F1 scores across all splits. For example, in the Education split, TabPFN attains an OOD accuracy of 0.800 and F1 of 0.724, outperforming both Random Forest (0.755, 0.625) and XGBoost (0.768, 0.660). Similar trends are observed in the Age and Gender splits. Notably, the F1 score improvements are substantial, which is particularly important for imbalanced classes. LinearSVC shows a marked drop in both accuracy and F1 in OOD settings, confirming its lack of robustness to distribution shift.

These results demonstrate that while tree-based models such as XGBoost and Random Forest remain highly competitive in in-distribution settings, TabPFN demonstrates superior robustness and generalization in OOD scenarios, especially in F1 score. This finding aligns with the hypothesis that meta-learning approaches, such as TabPFN, are better equipped to handle distribution shifts [9].

C. *Statistical Analysis of OOD Performance*

To assess whether the observed performance differences are statistically significant, paired t-tests were conducted and Wilcoxon signed-rank tests on the per-sample OOD accuracy and F1 scores. The objective of this statistical analysis is to determine whether TabPFN's improvements over baseline models are due to chance or represent a genuine advantage in OOD generalization.

The null hypothesis (H0) for each test is that there is no difference in mean performance (accuracy or F1) between TabPFN and the baseline model on the OOD test set. The alternative hypothesis (H1) is that there is a difference in mean performance. Two-tailed tests were used to ensure an unbiased evaluation, allowing for the possibility of both improvement and degradation relative to baselines. Both parametric (t-test) and non-parametric (Wilcoxon) tests were used to ensure robustness, as the distribution of per-sample scores may not be strictly normal [5].

To provide a compact summary of the statistical evidence across splits and baselines, Table V reports the p-values obtained from the paired tests for OOD accuracy and F1. For each comparison, lower p-values indicate stronger evidence against the null hypothesis of equal performance, while higher values suggest that any observed value may be attributable to sampling variability.

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TABLE V. STATISTICAL TEST RESULTS FOR OOD PERFORMANCE (P-VALUES)

Split	Comparison	Accuracy t-test	Accuracy Wilcoxon	F1 Wilcoxon
Education	TabPFN vs Random Forest	2.53e-05	2.71e-05	3.33e-165
	TabPFN vs XGBoost	0.0032	0.0032	3.33e-165
	TabPFN vs LinearSVC	1.46e-20	8.62e-20	3.33e-165
Age	TabPFN vs Random Forest	0.178	0.178	3.54e-165
	TabPFN vs XGBoost	0.495	0.495	7.82e-165
	TabPFN vs LinearSVC	4.34e-12	7.47e-12	3.33e-165
Gender	TabPFN vs Random Forest	9.81e-07	1.12e-06	3.33e-165
	TabPFN vs XGBoost	0.00044	0.00045	3.33e-165
	TabPFN vs LinearSVC	1.95e-10	2.87e-10	3.33e-165

As shown in Table V, the results show that TabPFN's OOD performance is statistically superior to both Random Forest and XGBoost in most cases, especially in F1 score. For the Education and Gender splits, the differences in both accuracy and F1 are highly significant ($p < 0.01$). For the Age split, the difference in accuracy is not statistically significant ($p > 0.05$), but the F1 difference remains highly significant ($p \approx 0$). This suggests that TabPFN is particularly effective at handling minority classes in OOD settings, which is reflected in the F1 metric. The use of both t-test and Wilcoxon tests confirms the robustness of these findings.

The choice of these statistical tests was motivated by the need to compare paired predictions on the same OOD samples. This approach ensured that the observed differences were not due to random variation. The consistent rejection of the null hypothesis in most cases supports the conclusion that TabPFN offers a genuine advantage in OOD generalization.

V. INTERPRETATION, LIMITATIONS, AND FUTURE WORK

The results of this study provide strong evidence that TabPFN is a state-of-the-art method for OOD generalization on tabular data. Its meta-learning approach, trained on a wide variety of synthetic tasks, enables it to adapt to new distributions more effectively than traditional models trained only on in-distribution data [3]. The improvements in F1 score are particularly important for real-world applications where class imbalance is common and minority class performance is critical [10].

Tree-based models such as Random Forest and XGBoost remain the best or highly competitive in in-distribution settings, but are generally outperformed by TabPFN in OOD scenarios. LinearSVC is not competitive in this context, especially under distribution shift.

The OOD splits based on education, age, and gender are realistic and challenging, making these results highly relevant for practical deployment in domains such as finance, healthcare, and social sciences, where demographic shifts are common [9].

Despite these strengths, there are limitations to this study. For some splits, such as Age, the accuracy difference between TabPFN and the baselines is not always significant, suggesting that the benefits of TabPFN may depend on the nature of the distribution shift. Further research is needed to explore more granular OOD scenarios, additional datasets, and potential hybrid approaches that combine the strengths of meta-learning and tree-based models.

Additionally, while TabPFN is evaluated in a zero-shot setting, the baselines are trained on in-distribution data, which may affect the fairness of the comparison. Future work should also consider the computational cost and scalability of TabPFN in larger tabular datasets.

Overall, TabPFN demonstrates robust and statistically significant improvements in OOD generalization for tabular data, particularly in F1 score. These findings highlight its potential for real-world applications where reliability and fairness under distribution shift are essential.

VI. CONCLUSION

This study evaluated the out-of-distribution (OOD) robustness of TabPFN, a meta-learning transformer model, on tabular data using the UCI Adult dataset. OOD splits were constructed based on education, age, and gender, and TabPFN was compared to established baselines, including Random Forest, XGBoost, and LinearSVC. The results show that TabPFN achieves superior generalization in OOD scenarios, particularly in terms of F1 score, which is crucial for imbalanced classification tasks.

Statistical analysis confirms that these improvements are highly significant in most cases. Tree-based models such as XGBoost and Random Forest remain highly competitive in in-distribution settings, but TabPFN consistently outperforms these baselines under distribution shift.

While the results are promising, few limitations should be acknowledged. For the Age split, the accuracy difference between TabPFN and the baselines is not always significant, suggesting that TabPFN's benefits may depend on the type of distribution shift. The comparison is asymmetric, as TabPFN leverages meta-learning while baselines are trained on in-distribution data.

Future research directions include evaluating TabPFN on additional tabular datasets with realistic demographic and domain-specific OOD splits to validate generalizability across diverse populations. Exploring hybrid approaches that combine TabPFN's meta-learning with tree-based models could leverage complementary strengths for improved performance-efficiency trade-offs.

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Despite these considerations, the findings provide clear and statistically significant evidence that TabPFN is a robust and effective method for OOD generalization on tabular data. This highlights its strong potential for real-world applications where reliability and fairness under distribution shift are essential.

REFERENCES

[1] V. Borisov, T. Leemann, K. Seßler, J. Haug, M. Pawelczyk, and G. Kasneci, "Deep neural networks and tabular data: A survey," IEEE Trans. Neural Netw. Learn. Syst., 2022, doi: 10.1109/TNNLS.2022.3229161.

[2] R. Shwartz-Ziv and A. Armon, "Tabular data: Deep learning is not all you need," Inf. Fusion, vol. 81, pp. 84–90, May 2022.

[3] N. Hollmann, S. Müller, K. Eggensperger, and F. Hutter, "TabPFN: A transformer that solves small tabular classification problems in a second," in Proc. Int. Conf. Learn. Represent. (ICLR), 2023.

[4] D. Dua and C. Graff, "UCI machine learning repository," University of California, Irvine, School of Information and Computer Sciences, 2017. [Online]. Available: http://archive.ics.uci.edu/ml

[5] [5] Student, "The probable error of a mean," Biometrika, vol. 6, no. 1, pp. 1–25, Mar. 1908

[6] F. Wilcoxon, "Individual comparisons by ranking methods," Biometrics Bull., vol. 1, no. 6, pp. 80–83, Dec. 1945.

[7] S. Popov, I. Morozov, and A. Babenko, "Neural oblivious decision ensembles for deep learning on tabular data," 2019, arXiv:1909.06312. [Online]. Available: https://arxiv.org/abs/1909.06312

[8] S. Arik and T. Pfister, "TabNet: Attentive interpretable tabular learning," in Proc. AAAI Conf. Artif. Intell., vol. 35, no. 8, 2021, pp. 6679–6687.

[9] J. Wang, C. Lan, C. Liu, Y. Ouyang, T. Qin, W. Lu, Y. Chen, W. Zeng, and P. Yu, "Generalizing to unseen domains: A survey on domain generalization," IEEE Trans. Knowl. Data Eng., vol. 35, no. 8, pp. 8052–8072, Aug. 2023.

[10] H. He and E. A. Garcia, "Learning from imbalanced data," IEEE Trans. Knowl. Data Eng., vol. 21, no. 9, pp. 1263–1284, Sep. 2009.

[11] K. Helli, D. Schnurr, N. Hollmann, S. Müller, and F. Hutter, "Drift-resilient TabPFN: In-context learning temporal distribution shifts on tabular data," in Proc. 38th Conf. Neural Inf. Process. Syst. (NeurIPS), 2024.

[12] J. Gardner, Z. Popovic, and L. Schmidt, "Benchmarking distribution shift in tabular data with TableShift," in Proc. 37th Conf. Neural Inf. Process. Syst. (NeurIPS), Datasets and Benchmarks Track, 2023.

[13] A. Ginanjar, X. Li, P. Singh, and W. Hua, "Representation learning on out of distribution in tabular data," in Proc. IEEE Int. Conf. Artif. Intell. Comput. Technol. (IAICT), 2025.

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